

EC466-966: Estimation and Inference in Econometrics

Exercise # 5: Differences-in-Differences

Background: Consider a policy intervention that was implemented in a treatment region in 2020, while a control region did not receive the intervention. You have panel data on unemployment rates for both regions from 2018 to 2022.

Let Y_{ct} denote the unemployment rate in city c at time t , where $c \in \{\text{Treatment, Control}\}$ and $t \in \{2018, 2019, 2020, 2021, 2022\}$. The observed unemployment rates (in percentages) are:

City	2018	2019	2020	2021	2022
Treatment	6.2	6.5	5.8	5.4	5.1
Control	5.8	6.0	6.3	6.5	6.7

1. Calculate the simple difference-in-differences (DiD) estimate of the treatment effect using only the pre-treatment year (2019) and the first post-treatment year (2020). Interpret your result.

- *ANS: The change in unemployment for Treatment is $5.8 - 6.5 = -0.7$, and for Control is $6.3 - 6.0 = 0.3$. The DiD estimate is $(-0.7) - (0.3) = -1.0$ percentage points. This suggests the policy intervention reduced unemployment by 1.0 percentage point in the treatment region relative to the counterfactual without intervention.*

2. Suppose we model the unemployment rate as:

$$Y_{ct} = \beta_t + \gamma_c + \delta D_{ct} + u_{ct} \quad (1)$$

where $D_{ct} = 1$ if city c is the treatment city and $t \geq 2020$, and $D_{ct} = 0$ otherwise.

- (a) Explain the meaning of each parameter (β_t , γ_c , δ) in the context of this application.

- *ANS: β_t represents time fixed effects (common time trends affecting both cities equally), γ_c represents city fixed effects (time-invariant differences between cities), and δ represents the causal effect of the policy intervention on unemployment.*
- (b) What key assumption about the conditional mean function $E(Y_{0ct}|c, t)$ does this specification impose?
- *ANS: The specification imposes the parallel trends assumption: in the absence of treatment, the difference in unemployment between treatment and control cities would remain constant over time. Formally, $E(Y_{0ct}|c, t) = \beta_t + \gamma_c$, which is additively separable in time and city effects.*
- (c) Show algebraically that the DiD estimator identifies δ under this specification.
- *ANS: Under the model, for Treatment pre-treatment (2019): $E[Y_{T,2019}] = \beta_{2019} + \gamma_T$; post-treatment (2020): $E[Y_{T,2020}] = \beta_{2020} + \gamma_T + \delta$. For Control in both periods: $E[Y_{C,2019}] = \beta_{2019} + \gamma_C$ and $E[Y_{C,2020}] = \beta_{2020} + \gamma_C$. The DiD estimator is:*

$$\begin{aligned}
DiD &= [E[Y_{T,2020}] - E[Y_{T,2019}]] - [E[Y_{C,2020}] - E[Y_{C,2019}]] \\
&= [(\beta_{2020} + \gamma_T + \delta) - (\beta_{2019} + \gamma_T)] - [(\beta_{2020} + \gamma_C) - (\beta_{2019} + \gamma_C)] \\
&= [\beta_{2020} - \beta_{2019} + \delta] - [\beta_{2020} - \beta_{2019}] = \delta
\end{aligned}$$

Thus, the DiD estimator identifies the treatment effect δ .

3. Using potential outcomes notation, let $Y_{ct}(1)$ denote the unemployment rate that would be observed in city c at time t if the treatment were implemented, and $Y_{ct}(0)$ the unemployment rate if the treatment were not implemented.

- (a) Write down the expression for the Average Treatment Effect on the Treated (ATT) at time $t = 2020$.
- *ANS: The ATT at $t = 2020$ is $ATT_{2020} = E[Y_{1,T,2020} - Y_{0,T,2020} | Treatment]$, where $Y_{1,T,2020}$ is the potential outcome with treatment and $Y_{0,T,2020}$ is the potential outcome without treatment for the treatment city.*
- (b) Explain why we cannot directly compute the ATT from observed data alone.
- *ANS: We cannot directly compute the ATT because we observe only $Y_{1,T,2020}$ (the actual treated outcome) but not $Y_{0,T,2020}$ (the counterfactual outcome without treatment). The fundamental problem of causal*

inference is that we cannot observe both potential outcomes for the same unit at the same time.

(c) State the parallel trends assumption formally using potential outcomes notation.

- *ANS: The parallel trends assumption states that $E[Y_{0,T,t} - Y_{0,T,t'}] = E[Y_{0,C,t} - Y_{0,C,t'}]$ for all t, t' . In other words, in the absence of treatment, the average change in outcomes over time would have been the same for both treatment and control groups.*